MING'S PERSONAL STUDY NOTES

Causal Inference for the Brave and True

Chapters 24–25 and Appendix

English translation of the handwritten notebook

How to read this edition

The translation follows the handwritten notes, including explanations, intuitions, questions, derivations, and marginal additions. Main pages and their supplements are joined by topic and by the arrows in the scans, rather than by a fixed odd/even page rule.

Scan references use one-based PDF page numbers. Author annotations preserve remarks visible in the notebook; they do not claim independent verification that a remark is absent from the textbook. Translator's notes identify uncertain handwriting or apparent source notation issues. They are separate from the author's notes. This is a translation, not a replacement textbook or a mathematical errata edition.

Source: `Chapter24-25.pdf`, 14 scanned pages. The accompanying source-order audit records page coverage and unresolved readings.

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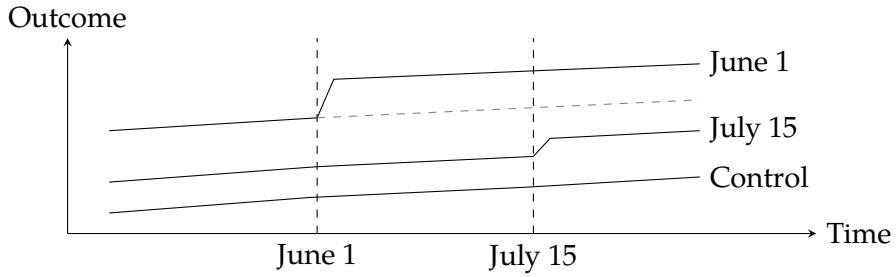
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Chapter 24: The Difference-in-Differences Saga

Panel data and two-way fixed effects

Scan reference: Chapter24-25.pdf, pp. 2 and 1; p. 1 continues the assumptions at the bottom of p. 2.

Panel data: repeatedly observing the same unit over time. The example compares units across time and across regions. The drawing has an early-treated group (June 1), a later-treated group (July 15), and a control group; a faint extra trajectory is labelled *TWFE prediction / counterfactual*.



Schematic reconstruction of the handwritten trajectories; no numerical data are implied.

Group units by their intervention date g (three groups in the example). The target written in the notes is

$$\text{ATT} = E[Y_1 \mid t > g] - E[Y_0 \mid t > g].$$

The model is

$$\text{Install}_{it} = \tau D_{it} + \gamma_i + \theta_t + e_{it}.$$

Here i denotes a unit, t time, and e_{it} noise; γ_i is associated with i and θ_t with t. Estimating this by OLS gives the **two-way fixed effects** (**TWFE**) model.

Author's annotations. At the same time, control for all characteristics. The red note connects TWFE and DiD: with two groups and two periods, this becomes DiD. The notes write the mathematically equivalent residualized regression as

$$\widetilde{\text{Install}}_{it} = \tau \widetilde{D}_{it} + e_{it},$$

with the following demeaning expressions:

$$\widetilde{\text{Install}}_{it} = \text{Install}_{it} - \underbrace{\frac{1}{T}\sum_{t=0}^{T} \text{Install}_{it}}_{\text{time average}} - \underbrace{\frac{1}{N}\sum_{i=0}^{N} \text{Install}_{it}}_{\text{unit average}},$$

$$\widetilde{D}_{it} = D_{it} - \frac{1}{T}\sum_{t=0}^{T} D_{it} - \frac{1}{N}\sum_{i=0}^{N} D_{it}.$$

Translator's note: These reproduce the scan, including summation endpoints. The usual double-demeaning expression also adds back the overall mean. This is a notation issue in the manuscript, not a change made in the translation.

Death: Failures over Effect Heterogeneity

TWFE often has bias when effects are heterogeneous. The fixed-effects model without time effects is written as

$$y_{it} = \tau D_{it} + \gamma_i + e_{it}.$$

The assumptions listed in the notebook are:

1. **Functional form:** no heterogeneous effect over time (constant effects); linearity in the covariates; additive fixed effects.
2. **Strict exogeneity:** parallel trends; no anticipation; no unobserved time-varying confounders; past treatment does not affect the current outcome (no carryover); past outcomes do not affect current treatment (no feedback).

The second item is the left-page continuation connected by the arrow at the foot of the main page.

Scan reference: Chapter24-25.pdf, p. 3.

Continuing the previous point: suppose the treatment effect is heterogeneous over time, and the treatment response is gradual (from $0.1, 0.2, \ldots$, increasing from day to day). The ATT estimated by TWFE is then biased. The drawing labels the early-treated group 1, the late-treated group 2, and the never-treated group 3. Decomposing TWFE into several 2×2 DiDs gives:

$$1-3:\ \checkmark, \quad 2-3:\ \checkmark, \quad 1-2:\ \checkmark, \quad 2-1:\ ??$$

The second member of each comparison is the control. **Author's intuition:** if the early-treated group 1 is used as the control, the model thinks that the untreated outcome Y_0 would have risen in parallel with that group's earlier trajectory. The purple curve in the drawing is this estimated Y_0, not the observed late-group outcome.

The notebook attributes this shorthand to Goodman-Bacon (2019):

$$\operatorname*{plim}_{n\to\infty} \widehat{\tau}^{\text{TWFE}} = \text{VWATT} - \Delta\text{ATT}.$$

The purple annotation interprets the bias term as the extent to which ATT changes over time. If the effect increases over time, as in the drawing, the bias is downward; in the opposite case it is upward. **Author's warning:** with negative effects that decay over time, even the estimated sign can be wrong. *Translator's note:* The limit subscript is indistinct; $n \to \infty$ renders the apparent probability-limit notation. VWATT is retained as written.

Event-study design

Rewrite TWFE using event-time indicators. The lead/lag expansion in the notes is rendered below with its uncertain final separator made explicit:

$$\begin{aligned} Y_{it} &= \tau^{-K} D_{ik}^{<-K} + \sum_{k=-K}^{-2} \tau^{\text{lead}} D_{it}^{k} + \sum_{k=0}^{L} \gamma_k^{\text{lag}} D_{it}^{k} \\ &\quad + \gamma_k^{L+} D_{it}^{>L} + \gamma_i + e_{it}, \\ D_{it}^{k} &= \mathbf{1}\{t - \text{Cohort}_i = k\}. \end{aligned}$$

The indicator is 1 if the relative period equals k, and 0 otherwise. **Author's annotation:** add a variable measuring how far a unit is from its treatment start date. This is labelled a *partial improvement*. *Translator's note:* The early-tail subscript looks like ik; the lower lead-sum bound is faint; the sign before γ_i is missing or unclear. The displayed equation uses $k = -K$ and a separating plus, but retains the manuscript's coefficient labels. No time fixed effect has been added.

Enlightenment: A Flexible Functional Form

Scan reference: Chapter24-25.pdf, p. 4.

A new model gives every individual a treatment effect at every time:

$$\text{Installs}_{it} = \sum_{i=0}^{N} \sum_{t=0}^{T} \tau_{it} D_{it} + \gamma_i + \theta_t + e_{it}.$$

There are too many parameters. Instead, assign an effect to each group g at each time:

$$\text{Installs}_{it} = \sum_{g=0}^{G}\sum_{t=0}^{T}\tau_{gt}D_{it} + \gamma_i + \theta_t + e_{it}.$$

This is marked as a better version. Reduce the parameters further by removing pre-treatment coefficients and the control-group coefficients:

$$\text{Installs}_{it} = \sum_{g=q}^{G}\sum_{t\geq g}^{T}\tau_{gt}D_{it} + \gamma_i + \theta_t + e_{it}.$$

Cohorts with $g < q$ are defined as controls. **Author's annotation:** one needs to create cohort dummy variables manually, such as $(0,1),(1,0),\ldots$. *Translator's note:* These sums reproduce the notebook's compact notation. Cohort-by-time indicators are not fully written out; the note about dummy variables explains the intended expansion. Remaining limitations:

1. Treatment effects can change as time passes; this was not considered before.
2. DiD and TWFE must satisfy the parallel-trends assumption. The restriction is very strong.

Chapter 25: Synthetic Difference-in-Differences

Representing panel data

Scan reference: Chapter24-25.pdf, p. 5.

There are multiple units over multiple periods, with treated and untreated observations mixed together. In D, **rows are times and columns are units**:

$$D = \begin{pmatrix} 0 & 0 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 0 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 1 & 1 \\ 0 & 0 & 0 & \cdots & 1 & 1 \end{pmatrix}.$$

The final columns identify the treated units. The note calls this *one-time treatment* and stars the design requirement: once a unit is treated, it does not return to an untreated state.

Representation 1: synthetic control. Collapse the display into four blocks:

$$D = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \qquad Y = \begin{pmatrix} Y_{\text{pre,co}} & Y_{\text{pre,tr}} \\ Y_{\text{post,co}} & Y_{\text{post,tr}} \end{pmatrix}.$$

Here pre means before treatment, post after treatment, co control group, and tr treatment group.

Representation 2: DiD. Use a long table with columns for unit, time, outcome, whether the unit belongs to the treatment group, and whether it has already been treated. The observed outcomes are

$$Y = \begin{pmatrix} Y(0)_{\text{pre,co}} & Y(0)_{\text{pre,tr}} \\ Y(0)_{\text{post,co}} & Y(1)_{\text{post,tr}} \end{pmatrix},$$

and

$$\text{ATT} = Y(1)_{\text{post,tr}} - \underbrace{Y(0)_{\text{post,tr}}}_{\text{missing}}.$$

Revisiting DiD and synthetic controls

Scan reference: Chapter24-25.pdf, pp. 6 and 7; p. 7 supplies the two comparison sketches.

The DiD regression is

$$Y_{it} = \beta_0 + \beta_1 \text{Post}_t + \beta_2 \text{Treated}_i + \beta_3 \text{Treated}_i \text{Post}_t + e_{it}.$$

Its assumption is

$$E[Y(0)_{\text{post,co}} - Y(0)_{\text{pre,co}}] = E[Y(0)_{\text{post,tr}} - Y(0)_{\text{pre,tr}}].$$

That is, in the absence of treatment the control and treatment groups have the same trend.

Rewrite the estimator in a form comparable to SDID:

$$\widehat{\tau}^{\text{did}} = \underset{\mu,\alpha,\beta,\tau}{\text{argmin}} \left\{ \sum_{i=1}^{N} \sum_{t=1}^{T} [Y_{it} - (\mu + \alpha_i + \beta_t + \tau D_{it})]^2 \right\}.$$

The notes connect this to demeaning:

$$\widetilde{Y}_{it} = Y_{it} - \overline{Y}_i - \overline{Y}_t = Y_{it} - T^{-1} \sum_{t=0}^{T} Y_{it} - N^{-1} \sum_{i=0}^{N} Y_{it},$$

$$\widetilde{D}_{it} = D_{it} - \overline{D}_i - \overline{D}_t = D_{it} - T^{-1} \sum_{t=0}^{T} D_{it} - N^{-1} \sum_{i=0}^{N} D_{it}.$$

This looks like synthetic controls, but lacks the weights. *Translator's note:* The arg min formulas use $\widehat{\tau}$ as shorthand for the treatment-coefficient component of the joint minimizer. The missing grand means are preserved from the scan.

For synthetic control, estimate weights using pre-treatment outcomes:

$$\widehat{w}^{\text{sc}} = \underset{w}{\text{argmin}} \, \|\overline{y}_{\text{pre,tr}} - Y_{\text{pre,co}} w_{\text{co}}\|_2^2, \qquad \sum_i w_i = 1, \quad w_i > 0 \quad \forall i.$$

Here $Y_{\text{pre,co}}$ is a $T_{\text{pre}} \times N_{\text{co}}$ outcome matrix: columns are units and rows are time periods. The vector w_{co} has size $N_{\text{co}} \times 1$. Each entry of $\overline{y}_{\text{pre,tr}}$ is the average of the treated units at a pre-treatment time.

Author's annotations. Fit a synthetic control to the treated group; this is similar to linear regression. After treatment, use the control outcomes and the fitted weights:

$$y_{\text{sc}} = Y_{\text{co}} \widehat{w}^{\text{sc}}.$$

This gives a good estimate of $Y(0)_{\text{post,tr}}$. Thus

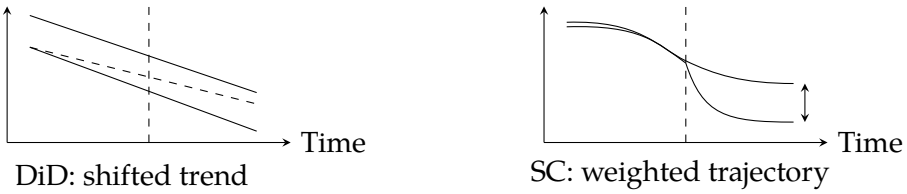
$$\widehat{\tau} = \overline{y}_{\text{post,tr}} - \overline{y}_{\text{post,sc}}.$$

Rewrite it as

$$\widehat{\tau}^{\text{sc}} = \underset{\beta,\tau}{\text{argmin}} \left\{ \sum_{i=1}^{N} \sum_{t=1}^{T} (Y_{it} - \beta_t - \tau D_{it})^2 \widehat{w}_i^{\text{sc}} \right\}.$$

The pre-treatment fitting step provides control-group weights; each treated unit has weight $1/N_{\text{tr}}$. Purple annotations emphasize: **no overall intercept μ, and no unit fixed effects α_i.**

Left-page visual intuition (p. 7). The DiD sketch shifts a trajectory into a parallel comparison; the handwritten comment says this amplifies the effect in the illustrated case. In the SC sketch, pre-treatment data generate weights and hence a synthetic untreated treatment group. Use the same weights to compute post-treatment outcomes, and estimate the effect from the difference between the two post-treatment values. The curves closely track before treatment, then separate.



Schematic reconstructions of the author's sketches.

Combining the two: synthetic DiD

Scan reference: Chapter24-25.pdf, pp. 8 and 10 (upper and middle portions).

The starred heading combines the synthetic-control weighted criterion and the DiD fixed-effects criterion:

$$\widehat{\tau}^{\text{sdid}} = \underset{\mu,\alpha,\beta,\tau}{\text{argmin}} \left\{ \sum_{i=1}^{N} \sum_{t=1}^{T} [Y_{it} - (\mu + \alpha_i + \beta_t + \tau D_{it})]^2 \underbrace{\widehat{w}_i^{\text{sdid}}}_{\text{unit weight}} \underbrace{\widehat{\lambda}_t^{\text{sdid}}}_{\text{time weight}} \right\}.$$

Time weights. Fit the average post-treatment outcome of the controls from their pre-treatment outcomes:

$$\widehat{\lambda}^{\text{sdid}} = \underset{\lambda}{\text{argmin}} \, \|\overline{y}_{\text{post,co}} - (\lambda_{\text{pre}} Y_{\text{pre,co}} + \lambda_0)\|_2^2, \qquad \sum_t \lambda_t = 1, \quad \lambda_t > 0.$$

- $Y_{\text{pre,co}}$ is $T_{\text{pre}} \times N_{\text{co}}$; rows are times and columns are units.
- $\overline{y}_{\text{post,co}}$ is a $1 \times N_{\text{co}}$ row vector. Each entry is a control unit's time-averaged outcome after treatment.
- λ_{pre} is a $1 \times T_{\text{pre}}$ row vector, with a weight for every pre-treatment period. λ is a time weight.

The arrows at the bottom of p. 8 show multiplication along the time dimension: prepend a row of ones to the matrix for the intercept, and fit $\overline{y}_{\text{post,co}}$ from the weighted pre-treatment rows to obtain λ_t. *Translator's note:* The constraint looks like $\lambda_t = 1$ without a visible summation sign; the displayed sum-to-one reading follows the adjacent weighting construction. Several explanatory lines at the bottom are crossed out and replaced by this matrix sketch.

Unit weights. The small diagram at the top of p. 10 connects control/pre to control/post by λ, and control/pre to treated/pre by w, giving the corresponding treated/post comparison.

$$\begin{aligned} \widehat{w}^{\text{sdid}} = \underset{w}{\text{argmin}} \; & \{\|\overline{y}_{\text{pre,tr}} - (Y_{\text{pre,co}} w_{\text{co}} + w_0)\|_2^2 \\ & + \zeta^2 T_{\text{pre}} \|w_{\text{co}}\|_2^2\}, \qquad \sum_i w_i = 1, \quad w_i > 0, \\ \zeta &= (N_{\text{tr}} T_{\text{post}})^{1/4} \widehat{\sigma}(\Delta_{it}), \\ \Delta_{it} &= Y_{it} - Y_{i,t-1}. \end{aligned}$$

Here Δ_{it} is the first difference and $\widehat{\sigma}(\Delta_{it})$ its standard deviation.

Author's annotations. The L_2 penalty makes the nonzero weights more widely distributed. Compared with the previous SC, this version (1) has an intercept, and (2) uses an L_2 penalty. The implementation instruction says to add a row of ones *before* $\overline{y}_{\text{pre,tr}}$ as an intercept and perform the regression. *Translator's note:* This instruction is reproduced from the scan; the displayed model places the intercept in the predictors. The scan again abbreviates the constraint as $w_i = 1$; it is read here as sum-to-one, consistently with p. 6.

For the treated units and post-treatment times, use uniform weights $1/N_{\text{tr}}$ and $1/T_{\text{post}}$, respectively. **Author's intuition:** compared with SC, SDID pays attention only to data that are close to the expected treatment effect.

Time-effect heterogeneity and staggered adoption

Scan reference: Chapter24-25.pdf, p. 10 (lower portion), then the p. 9 supplement.

SDID makes it easy to estimate an effect for each period: simply run the procedure multiple times. In these handwritten matrices, subscripts on the zeros and ones identify the period.

Time-effect heterogeneity. The example is

$$D = \begin{pmatrix} 0_1 & 0_1 & 0_1 \\ 0_2 & 0_2 & 0_2 \\ 0_3 & 0_3 & 1_3 \\ 0_4 & 0_4 & 1_4 \end{pmatrix}.$$

Applying SDID gives an average ATT over periods 3 and 4. To estimate these periods separately, use

$$D_1 = \begin{pmatrix} 0_1 & 0_1 & 0_1 \\ 0_2 & 0_2 & 0_2 \\ 0_3 & 0_3 & 1_3 \end{pmatrix}, \qquad D_2 = \begin{pmatrix} 0_1 & 0_1 & 0_1 \\ 0_2 & 0_2 & 0_2 \\ 0_4 & 0_4 & 1_4 \end{pmatrix}.$$

Staggered adoption: left-page supplement. The notebook gives

$$D = \begin{pmatrix} 0_1 & 0_1 & 0_1 \\ 0_2 & 0_2 & 0_2 \\ 0_3 & 0_3 & 1_3 \\ 0_4 & 1_4 & 1_4 \end{pmatrix}.$$

The first decomposition is

$$D_1 = \begin{pmatrix} 0_1 & 0_1 & 0_1 \\ 0_2 & 0_2 & 0_2 \\ 0_4 & 1_4 & 1_4 \end{pmatrix}, \qquad D_2 = \begin{pmatrix} 0_1 & 0_1 & 0_1 \\ 0_2 & 0_2 & 0_2 \\ 0_3 & 0_3 & 1_3 \end{pmatrix}.$$

Weight the ATT from D_1 by $2/3$, and the ATT from D_2 by $1/3$. Alternatively, split by treated cohort:

$$D_1 = \begin{pmatrix} 0_1 & 0_1 \\ 0_2 & 0_2 \\ 0_3 & 0_3 \\ 0_4 & 1_4 \end{pmatrix}, \qquad D_2 = \begin{pmatrix} 0_1 & 0_1 \\ 0_2 & 0_2 \\ 0_3 & 1_3 \\ 0_4 & 1_4 \end{pmatrix}.$$

The example underneath says: California, treated for 12 years; others, 8 years $\times 3$. The weighting calculation is

$$\text{ATT}_0 \frac{12}{36} + \left(\sum \text{ATT}\right) \frac{1}{3} \frac{3 \times 8}{36}.$$

This preserves the author's treatment-duration weighting illustration.

Placebo variance estimation

Scan reference: Chapter24-25.pdf, p. 11.

Compared with SC, the notebook describes SDID as having lower error bars. SDID captures variation in outcomes over time and across units. This improves the SDID confidence interval and applies to a single treated unit.

For each control unit, pretend it received treatment and use SDID to estimate a placebo ATT. The notes write

$$\widehat{V}_{\tau}^{\text{placebo}} = B^{-1} \sum_{b=1}^{B} \left(\widehat{\tau}^{(b)} - \overline{\tau}^{(b)} \right)^2, \qquad \tau \in \widehat{\tau}^{\text{sdid}} \pm z_{\alpha/2} \sqrt{\widehat{V}_{\tau}}.$$

The variance of these estimates is the SDID placebo variance; use it to construct the confidence interval. A parenthetical annotation says this is the same as SC. *Translator's note:* The superscript on the handwritten mean $\overline{\tau}$ looks like (b), and the normal critical value is written $z_{\alpha/2}$. Both are retained. The intended placebo mean and quantile convention are not explicitly defined.

Appendix: Debiasing and Prediction

Scan reference: Chapter24-25.pdf, p. 12.

A heading note links the treatment of non-randomized data to **propensity scores** and **orthogonalization**.

Debiasing with orthogonalization

The purple annotation refers to Chapter 22. All features are contained in X.

1. Take the first group of features and fit

   $$y_i = \theta_0 + \theta_1 X_{1i} + e_i.$$

   Obtain the residual

   $$y_i^* = y_i - (\widehat{\theta}_0 + \widehat{\theta}_1 X_{1i}).$$

2. Use the first group of features to predict the second group:

   $$X_{2i} = \gamma_0 + \gamma_1 X_{1i} + e_i.$$

   Obtain the residual

   $$X_{2i}^* = X_{2i} - (\widehat{\gamma}_0 + \widehat{\gamma}_1 X_{1i}).$$

3. Regress the outcome residual on the feature residual:

   $$y_i^* = \beta_0 + \beta_2 X_{2i}^* + e_i.$$

Debiasing with propensity scores

Scan reference: Chapter24-25.pdf, p. 13.

The purple annotation refers to Chapter 11. Weight each unit. For binary treatment,

$$w_i = \frac{T}{P(T = 1 \mid X)} + \frac{1 - T}{1 - P(T = 1 \mid X)}.$$

The next formula, labelled *continuous* in the manuscript, is

$$w_i = \sum_{k=0}^{K} \frac{\mathbf{1}\{T = k\}}{P(T = k \mid X)}.$$

Translator's note: The label is translated literally; the sum is a discrete, multiple-treatment-level expression.

Author's intuition. A unit with a high probability of treatment that nevertheless remains untreated essentially resembles a treated unit but is observed in the control group. It can be used to estimate $Y_0 \mid T = 1$. If propensity scores are unavailable, estimate them using the probability of receiving treatment as the target, with machine learning. A small numerical illustration mentions 20%, 0.2, and receiving an email marked 1; the rest of that short line is unclear in the scan.

Weaknesses of propensity scores.

1. When the propensity score is very small ($e < 0.05$), the weight is too large: the observation effectively appears too many times and strongly influences the debiased dataset. The note mentions manual trimming. The purple annotation stresses **high variance**.
2. **Positivity or common support:** there must be enough overlap between the characteristics of treated and control units. If overlap is too limited, propensity-score adjustment does a poor job of removing the association.

The red annotations add: manual intervention is not very advisable because it can introduce excessive subjective bias; do not extrapolate to population subsets where positivity does not hold.

When prediction fails

Scan reference: Chapter24-25.pdf, p. 14.

Prediction and causality. This is a heading-only page; no further notes are written beneath it.